General Education Reflection

Over my time at Iowa State, I have taken the opportunity to explore subjects outside my primary engineering field. Although my major has taught me how to approach technical problems and develop solutions, my general education courses have helped me know the broader context in which those solutions occur. These classes introduced me to viewpoints and insights that I would not have encountered through engineering classes alone. I learned that an effective engineering solution requires more than technical knowledge; it also requires understanding the people affected, the community's needs, and the broader social environment in which it will be implemented. Additionally, interacting with students from different academic backgrounds has helped me become a better communicator. I have learned to explain technical concepts in ways that are accessible to people without an engineering background while appreciating their own knowledge and experiences. Looking back, my general education courses have contributed to my maturation not only as a student, but also as a person who can handle problems with a more open mind.

One course that contributed to this growth was my American Indian Studies course. This class challenged me to evaluate critically the consequences of maintaining harmful systems and refusing to reconsider established beliefs. Through examining the history of Native American communities, I gained a better understanding of how decisions made by those in positions of power can have lasting effects on entire populations. Those historical events also encouraged me to think about how similar patterns can appear in modern society. For example, talks surrounding immigration enforcement and the use of law enforcement against people considered "undesirable" raised questions about how people in power determine who belongs and how those decisions affect people and communities. Although the historical circumstances differ, examining these issues helped me more deeply understand the human consequences of policies and institutions. This perspective connects to engineering because technological solutions do not exist independently of society. Engineers must consider who benefits from a solution, who might be negatively affected, and whether the systems they develop solve a problem or unintentionally reinforce ongoing inequalities.

My military history course continued to develop this way of thinking, although it approached the subject from a different perspective. While my American Indian Studies course encouraged me to examine the consequences of social and political decisions, my military history course explored the challenges, mistakes, and decisions behind past military conflicts. Studying these events furthered my understanding that learning history involves more than memorizing dates, battles, or important figures. It also requires looking into the decisions they made, what information they had available, and how their choices shaped the futures of their kingdoms and peoples. I found it fun that, even when studying events that have already occurred, there are still opportunities to ask so many questions and build a deeper understanding of the

subject. This taught me that education does not have to be limited to the material directly presented in a classroom. We can keep making time for our own interests, especially when they give us new perspectives on problems. In engineering, this mindset is valuable because solving a technical problem often requires investigating previous attempts, understanding why they succeeded or failed, and using that knowledge to develop a better approach.

While these two courses helped me think more critically about society and the decisions people make, my wine and spirits course helped me apply what I learned in a more personal and social setting. Before taking this class, I didn't fully appreciate the history, knowledge, and attention to detail involved in winemaking. The lectures introduced me to the processes and factors that influence the final product, giving me a stronger appreciation for the work behind something I had previously only scratched the surface of. Beyond the course material, the class also provided opportunities to participate in small discussions and interact with other students. These conversations helped me become more comfortable connecting with people who had different interests and experiences from my own. I also learned that communication is about identifying common ground, listening to others, and learning from their perspectives. These skills are especially important on engineering teams, where people may approach the same problem with different levels of experience and different ideas about how to solve it. Clear communication, understanding others, and building relationships make collaboration more effective and help a team work toward a shared goal.

Together, these courses have shown me that personal and career development requires more than developing expertise in a single subject. My American Indian Studies course encouraged me to question established systems and consider the human consequences of decisions. My history course taught me to learn from past mistakes and approach problems with an open mind rather than presumptions. My wine and spirits course helped me build my communication skills and interact with people from different backgrounds. Although each course taught me differently, they all pushed me to look beyond my immediate perspective and consider information I might otherwise miss.

To conclude, my time at Iowa State and the general education courses I have taken have helped me grow into a better student, communicator, and engineer. These experiences have shown me that technical knowledge is most valuable when combined with an understanding of the people and communities it serves. As I go on this path, I want to carry these lessons forward by understanding not only how to recognize vulnerabilities and develop effective defenses, but also how security decisions affect the people who rely on the systems being protected. Overall, my general education courses have helped me understand that success in my job involves more than solving technical problems. It also means recognizing different perspectives, learning from the past, communicating effectively, and using my knowledge in ways that benefit others.